

Indian Institute of Technology Dharwad

CS201L: Artificial Intelligence Laboratory

Lab 3: Data Cleaning - Handling Missing Values - Practice Questions

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Practice Problem Statement

A real-world medical dataset containing sales records is provided in a CSV file (`sales_data.csv`).

Write a Python program to perform the following tasks on the given dataset:

1. Display the number of missing values in each attribute. Also, find the total number of missing values in the file.
2. Analyze the missing values in `sales_data.csv` by counting the number of tuples having one to thirteen missing values and by identifying tuples with at least 50% missing attributes. Plot a graph showing the number of missing values versus the number of tuples.
3. Clean the dataset by removing tuples having 50% or more missing attributes and tuples with missing values in the target attribute **Revenue**. Save the cleaned dataset as `sales_data_updated.csv`.
4. For the file `sales_data_updated.csv`, compute the mean, median, mode, and standard deviation for all numerical attributes before handling missing values.
5. Handle missing values in numerical attributes using the following techniques: **Median Imputation, Forward Fill, Backward Fill, and Linear Interpolation**. After each technique, count and display the number of missing values in each attribute.
6. After replacing missing values using each technique, recompute the mean, median, mode, and standard deviation for all numerical attributes and analyze the changes in the statistical measures.